

Preliminary Concepts

Lesson 1.1: Binary Operations

Learning Outcomes: At the end of the lesson, the students are able to

- verify if a given mapping is a well-defined binary operation;
- test for commutativity and associativity of a given binary operation;
- check the existence of identity elements and inverse of elements;
- construct and interpret Cayley tables for binary operations on finite sets.

What is a Binary Operation?

What do we mean when we say that $3 + 5 = 8$? In general, what do we mean by *addition*? Regardless of its meaning, addition may be considered as a process where we take two numbers as inputs and returns a uniquely determined number as an output. As in the chosen example above, 3 and 5 are the inputs and 8 is the output.

Mathematically speaking, we may think of addition (+) as a mapping or function that assigns to every ordered pair (a, b) of numbers a unique number which we denote as $a + b$. The following definition generalizes the idea.

Definition 1.1.1 (Binary Operation)

A binary operation on a nonempty set S is any mapping $*$: $S \times S \rightarrow S$. If $(a, b) \in S \times S$ and $*((a, b)) = c$, we write $a * b = c$.

To define a binary operation on a set S , we must specify for each ordered pair $(a, b) \in S \times S$ a unique element of S . However, some well-known binary operations in mathematics are already pre-defined.



Example 1.1.2 Addition (+), subtraction (−) and multiplication (·) are binary operations on the set $\mathbb{R}$ of real numbers. However, division (÷) is **not** a binary operation on $\mathbb{R}$ since any number divided by 0 is not defined. This would mean that ÷ does not return an output for an ordered pair of the form $(x, 0)$, for any real number x .

Remark 1.1.3 If $*$ is a binary operation on a set S , then for all $a, b \in S$, $a * b \in S$. In other words, we must ensure that the result of *performing the operation* to every pair of elements must be defined in the same set.

Example 1.1.4 Define the operation $\boxtimes$ on the set $\mathbb{Z}$ of integers as follows:

$$a \boxtimes b := a^2 + b^2, \quad \forall a, b \in \mathbb{Z}.$$

We may observe that $\boxtimes$ is a binary operation on $\mathbb{Z}$ since $a^2 + b^2 \in \mathbb{Z}$, whenever $a, b \in \mathbb{Z}$. Let us calculate the expressions $(2 \boxtimes 3) \boxtimes 4$ and $2 \boxtimes (3 \boxtimes 4)$. The pairs of parentheses indicate which pair should be evaluated first. Note that $2 \boxtimes 3 = 2^2 + 3^2 = 13$. Thus,

$$(2 \boxtimes 3) \boxtimes 4 = 13 \boxtimes 4 = 13^2 + 4^2 = 185.$$

On the other hand, $(3 \boxtimes 4) = 3^2 + 4^2 = 25$, so

$$2 \boxtimes (3 \boxtimes 4) = 2 \boxtimes 25 = 2^2 + 25^2 = 229.$$

As we can see, $(2 \boxtimes 3) \boxtimes 4 \neq 2 \boxtimes (3 \boxtimes 4)$. This example shows that the role of grouping symbols is essential in conveying the correct meaning of an expression.

Example 1.1.5 Let X be a set and let $\mathcal{P}(X)$ be its power set (set of subsets). The set union ($\cup$) and set intersection ($\cap$) are binary operations on $\mathcal{P}(X)$. This is because of the fact that $A, B \in \mathcal{P}(X)$ implies that $A \cup B \in \mathcal{P}(X)$ and $A \cap B \in \mathcal{P}(X)$.

Example 1.1.6 Let X be a nonempty set and let $\mathcal{F}(X)$ denote the set of all functions from X to X . If $f : X \rightarrow X$ and $g : X \rightarrow X$ are functions, then their *composition* $f \circ g : X \rightarrow X$ defined by

$$(f \circ g)(x) = f(g(x)), \quad \forall x \in X,$$

is an element of $\mathcal{F}(X)$. See [Figure 1](#). Consequently, $\circ$ is a binary operation on $\mathcal{F}(X)$.



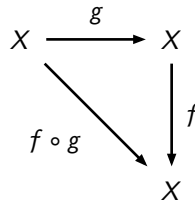


Figure 1: Function Composition

Example 1.1.7 (Linear Algebra Required) Let $M_2(\mathbb{R})$ denote the set of all 2-by-2 matrices with real entries. The following are well-known binary operations on $M_2(\mathbb{R})$.

1. *Matrix Addition*

$$\begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix} + \begin{pmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{pmatrix} = \begin{pmatrix} a_{11} + b_{11} & a_{12} + b_{12} \\ a_{21} + b_{21} & a_{22} + b_{22} \end{pmatrix}.$$

2. *Matrix Multiplication*

$$\begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix} \begin{pmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{pmatrix} = \begin{pmatrix} a_{11}b_{11} + a_{12}b_{21} & a_{11}b_{12} + a_{12}b_{22} \\ a_{21}b_{11} + a_{22}b_{21} & a_{21}b_{12} + a_{22}b_{22} \end{pmatrix}.$$

Note that the definitions of these binary operations may be extended to the set $M_n(\mathbb{R})$ of n -by- n real matrices.

Binary operations may be defined on an arbitrary set and may not be as *meaningful* as the previous examples. If $S = \{a_1, a_2, \dots, a_n\}$ is a finite set with n elements, we may define a binary operation $*$ by defining an n -by- n matrix $[p_{ij}]$, where

$$p_{ij} = a_i * a_j,$$

for all i and for all j . We normally present this definition of $*$ in tabular form as shown in Table 1.

$*$	a_1	a_2	$\dots$	a_j	$\dots$	a_n
a_1	p_{11}	p_{12}	$\dots$	p_{1j}	$\dots$	p_{1n}
a_2	p_{21}	p_{22}	$\dots$	p_{2j}	$\dots$	p_{2n}
$\vdots$	$\vdots$	$\vdots$		$\vdots$		$\vdots$
a_i	p_{i1}	p_{i2}	$\dots$	p_{ij}	$\dots$	p_{in}
$\vdots$	$\vdots$	$\vdots$		$\vdots$		$\vdots$
a_n	p_{n1}	p_{n2}	$\dots$	p_{nj}	$\dots$	p_{nn}

Table 1: Cayley Table

Example 1.1.8 Let $\odot$ be the binary operation on the set $S = \{1, 2, 3\}$ defined by the Cayley table below. Calculate $3 \odot ((2 \odot 1) \odot 3)$.



$\odot$	1	2	3
1	1	3	2
2	2	1	3
3	3	2	1

Solution. Starting with the innermost pair, note that

- $2 \odot 1 = 2$;
- $(2 \odot 1) \odot 3 = 2 \odot 3 = 3$;
- Therefore, $3 \odot ((2 \odot 1) \odot 3) = 3 \odot 3 = 1$.

How many distinct binary operations may be defined on a set with exactly n elements?

To answer this question, think of a binary operation on a set S with n elements as a Cayley table. To fill up the entire Cayley table, one must specify an element of S to each of the n^2 entries in the table. Since we may assign any of the n elements of S to each of these entries, the *fundamental counting principle* guarantees that there are exactly

$$\underbrace{n \cdot n \cdot \cdots \cdot n}_{n^2} = n^{n^2}$$

such binary operations.

We next look at how a binary operation behaves on a particular subset of a set. Consider the standard addition (+) on the set $\mathbb{R}$ and the subset $\mathbb{Z} \subseteq \mathbb{R}$. Note that if $a, b \in \mathbb{Z}$, then we know that $a + b \in \mathbb{Z}$. This means that the standard addition of real numbers may be considered as a binary operation on the set of integers as well.

Definition 1.1.9 Let $*$ be a binary operation on S and let H be a nonempty subset of S . We say that H is **closed** under $*$ if $a * b \in H$ whenever $a, b \in H$

Example 1.1.10 The following subset of $\mathbb{R}$ is closed under addition:

- $\mathbb{Q}$ (set of rational numbers)
- $\mathbb{Z}$ (set of integers)



- $2\mathbb{Z} = \{2k \mid k \in \mathbb{Z}\}$ (set of even integers)
- $\mathbb{N}$ (set of natural numbers)

However, the set $\mathbb{P}$ of all prime numbers, for example, is **not** closed under $+$ since $3 + 5 = 8$ is not prime, although both 3 and 5 are prime.

Example 1.1.11 Show that the set $H = \{a + b\sqrt{2} \mid a, b \in \mathbb{Q}\}$ is closed under the standard real number multiplication.

Proof. Let $h_1, h_2 \in H$. Then we may write

$$h_1 = a_1 + b_1\sqrt{2} \quad \text{and} \quad h_2 = a_2 + b_2\sqrt{2},$$

where $a_1, a_2, b_1, b_2 \in \mathbb{Q}$. Now,

$$\begin{aligned} h_1 h_2 &= (a_1 + b_1\sqrt{2})(a_2 + b_2\sqrt{2}) \\ &= (a_1 a_2 + 2b_1 b_2) + (a_1 b_2 + b_1 a_2)\sqrt{2}. \end{aligned}$$

Since $\mathbb{Q}$ is closed under the standard addition and multiplication, it follows that $a_1 a_2 + 2b_1 b_2 \in \mathbb{Q}$ and $a_1 b_2 + b_1 a_2 \in \mathbb{Q}$. Therefore, $h_1 h_2 \in H$, proving that H is closed under multiplication. $\square$

Properties of Binary Operations

From elementary mathematics, we know as an axiom that for all $a, b \in \mathbb{R}$,

$$a + b = b + a.$$

On the other hand, if we consider the functions $f : \mathbb{R} \rightarrow \mathbb{R}$ and $g : \mathbb{R} \rightarrow \mathbb{R}$ given by

$$f(x) = x + 1 \quad \text{and} \quad g(x) = 2x - 3,$$

then

$$(f \circ g)(x) = f(g(x)) = f(2x - 3) = (2x - 3) + 1 = 2x - 2,$$

while

$$(g \circ f)(x) = g(f(x)) = g(x + 1) = 2(x + 1) - 3 = 2x - 1.$$

In particular, $(f \circ g)(0) = -2 \neq -1 = (g \circ f)(0)$. Therefore,



$$f \circ g \neq g \circ f.$$

This shows that the fact that $a + b = b + a$, for all $a, b \in \mathbb{R}$ is a special occasion.

Definition 1.1.12 A binary operation $*$ on S is said to be

1. **commutative** if for all $a, b \in S$, $a * b = b * a$.
2. **associative** if for all $a, b, c \in S$, $a * (b * c) = (a * b) * c$.

Example 1.1.13 The standard addition and standard multiplication on $\mathbb{R}$ are both commutative and associative. These facts are axioms for the real number system. However, real number subtraction is neither commutative nor associative. Can you think of a counterexample?

Example 1.1.14 Consider the binary operation $\boxtimes$ defined on $\mathbb{Z}$ by $a \boxtimes b = a^2 + b^2$. Note that $\boxtimes$ is commutative since for all $a, b \in \mathbb{Z}$, we have

$$a \boxtimes b = a^2 + b^2 = b^2 + a^2 = b \boxtimes a.$$

However, in a previous example, we have shown that $2 \boxtimes (3 \boxtimes 4) \neq (2 \boxtimes 3) \boxtimes 4$. Therefore, $\boxtimes$ is **not** associative.

Example 1.1.15 Function composition on $\mathcal{F}(X)$ is associative but not commutative, in general. To see this, let $x \in X$ and let $f, g, h \in \mathcal{F}(X)$. Then

$$\begin{aligned} [f \circ (g \circ h)](x) &= f((g \circ h)(x)) = f(g(h(x))) \\ &= (f \circ g)(h(x)) = [(f \circ g) \circ h](x). \end{aligned}$$

Therefore, $f \circ (g \circ h) = (f \circ g) \circ h$, proving the function composition is associative. The introduction of [Section 1.1.2](#) exhibited two functions $f, g \in \mathcal{F}(\mathbb{R})$ such that $f \circ g \neq g \circ f$. This shows that function composition is **not** commutative in general.

Remark 1.1.16 Let $*$ be a commutative (associative) binary operation on S . If $H \subseteq S$ is closed under $*$, then $*$ is a commutative (resp. associative) binary operation on S .

Suppose that a binary operation $*$ is defined on a finite set $S = \{a_1, a_2, \dots, a_n\}$ using a Cayley table, whose (i, j) entry is p_{ij} . If $*$ is commutative, then



$$p_{ij} = a_i * a_j = a_j * a_i = p_{ji}.$$

This means that the Cayley table is *symmetric* with respect to its main diagonal.

Example 1.1.17 Let $*$ be the binary operation defined on $S = \{a, b, c\}$ by Table 2. Since the table is symmetric with respect to the main diagonal, the binary operation $*$ is commutative. Corresponding elements are highlighted using the same color.

$*$	a	b	c
a	a	b	c
b	b	c	a
c	c	a	b

Definition 1.1.18 (Identity Element)

Let $*$ be a binary operation on S . We say that an element $e \in S$ is an **identity element** under $*$

$$a * e = a = e * a, \text{ for all } a \in S.$$

Notice that in Definition 1.1.18, we refer to an identity element using the indefinite article “an”. We ask: is it possible for a set to have two identity elements under a given binary operation?

Let $*$ be a binary operation on S . Suppose that e_1 and e_2 are identity elements of S . By definition, $e_1 * a = a$, for all $a \in S$. In particular, by setting $a = e_2$ we have

$$e_1 * e_2 = e_2. \quad (1)$$

Since e_2 is also an identity element, it follows that $a * e_2 = a$, for all $a \in S$. By setting $b = e_1$ we have

$$e_1 * e_2 = e_1. \quad (2)$$

Combining Equation 1 and Equation 2, we get $e_1 = e_1 * e_2 = e_2$. We have proven the following result.

Theorem 1.1.19 If $*$ is a binary operation on S , then S can have at most one identity element under $*$.

When referring to an identity element, we may now use the definite article “the” since we cannot have two identity elements in a given set under a particular binary operation.



Example 1.1.20 The real number 0 is the identity element of $\mathbb{R}$ under the standard addition, while the real number 1 is an identity element of $\mathbb{R}$ under the standard multiplication.

Example 1.1.21 Let X be a set. Note that

- for all $A \in \mathcal{P}(X)$, we have $A \cup \emptyset = A$. Thus, the empty set $\emptyset$ is the identity element of $\mathcal{P}(X)$ under set union.
- for all $A \in \mathcal{P}(X)$, we have $A \cap X = A$. Thus, X is the identity element of $\mathcal{P}(X)$ under set intersection.

Example 1.1.22 Let X be a nonempty set. Define the function $\text{id}_X : X \rightarrow X$ as

$$\text{id}_X(x) = x, \text{ for all } x \in X.$$

If $f \in \mathcal{F}(X)$, then for all $x \in X$

$$(f \circ \text{id}_X)(x) = f(\text{id}_X(x)) = f(x),$$

and that

$$(\text{id}_X \circ f)(x) = \text{id}_X(f(x)) = f(x).$$

Therefore, $f \circ \text{id}_X = f = \text{id}_X \circ f$ ($\forall f \in \mathcal{F}(X)$) making the function id_X the identity element of $\mathcal{F}(X)$ under composition of functions. For this reason, the function id_X is called as the **identity function** on X .

Definition 1.1.23 Let $*$ be a binary operation on S , with identity element e . Let $a \in S$. We say that

1. $b \in S$ is a **left-inverse** of a if $b * a = e$.
2. $b \in S$ is a **right-inverse** of a if $a * b = e$.
3. $b \in S$ is an **inverse** of a if b is both a left-inverse and a right-inverse of a .

An element with an inverse is said to be **invertible**.

Example 1.1.24 In $\mathbb{R}$, we know from elementary algebra that

- every element a has an inverse under addition, namely $-a$ (additive inverse). Note that, $a + (-a) = 0$.
- every **nonzero** element a has an unverse under multiplication, namely $\frac{1}{a}$ (multiplicative inverse). Note that $a \cdot \frac{1}{a} = 1$. ($a \neq 0$)

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Example 1.1.25 Consider the binary operation $\otimes$ on the set $L = \{1, 2, 3, 4, 5\}$ given by Table 3.

$\otimes$	1	2	3	4	5
1	1	2	3	4	5
2	2	3	1	5	4
3	3	4	5	1	2
4	4	5	2	3	1
5	5	1	4	2	3

Table 3: Cayley Table

element	left-inverse	right-inverse	inverse
1	1	1	1
2	5	3	none
3	2	4	none
4	3	5	none
5	4	2	none

Table 4: Inverses of Elements

Notice that 1 is the identity element of L under $\otimes$. Table 4 summarizes the left-inverse, right-inverse, and inverse of each element in L . For example, consider the element 2. looking at the second column of Table 3, we see that $5 * 2 = 1$, hence 5 is a left-inverse of 2. On the other hand, if we look at the second row of Table 3, we see that $2 * 3 = 1$, so 3 is a right-inverse of 2. Although, 2 has a left-inverse and a right-inverse, 2 is **not** invertible because its left-inverse and right-inverse are not equal. We can perform this check to each element of L to see that only the identity element 1 is invertible.

Note In general, a left-inverse and a right-inverse for an element may not be equal. For an element to be invertible, it must have the same left-inverse and right-inverse. In this case, the common element is an inverse of the given element also known as a *two-sided inverse*.

We next show how associativity can affect the behavior of left and right-inverses.

Theorem 1.1.26 Let $*$ be an associative binary operation on a set S , with an identity element e . If an element $a \in S$ has a left-inverse ℓ and has a right-inverse r , then $\ell = r$. Consequently, every invertible element has a unique inverse.

Proof. Let $a \in S$. Suppose that a has a left-inverse ℓ and a right-inverse r . Then $\ell * a = e$ and $a * r = e$. From these equations, it follows that

$$\ell = \ell * e = \ell * (a * r) = (\ell * a) * r = e * r = r.$$

Now, let $a \in S$ be invertible. Suppose that b_1 and b_2 are inverses of a . Then b_1 and b_2 are both left-inverses and right-inverses of a . By the first part, $b_1 = b_2$. This completes the proof of the theorem. $\square$